

Computer Networks Labsheet2

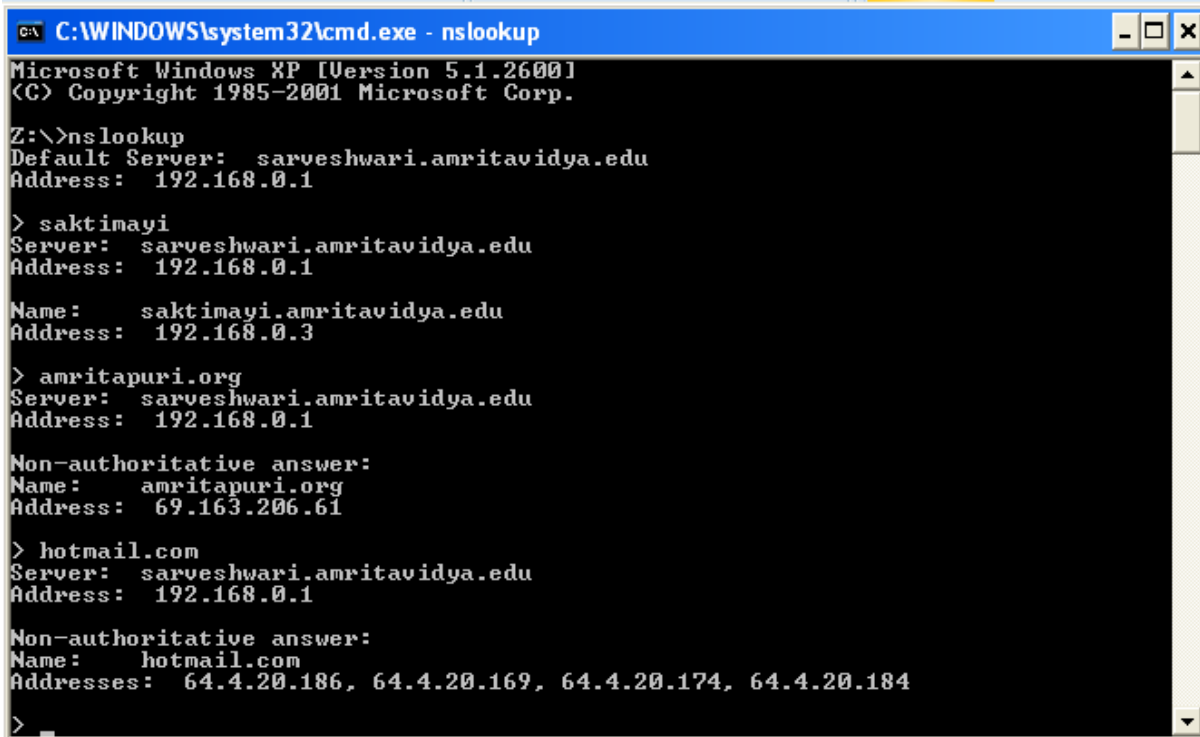
Analyzing the Web pages accessed using Wireshark

Part1: Mapping the given URL to IP address using nslookup

To run *nslookup* in Linux/Unix, you just type the *nslookup* command on the command line. To run it in Windows, open the Command Prompt and run *nslookup* on the command line.

We as humans usually use the website address known as Uniform Resource Locator (URL) to identify the destination host. But IP address is used by the devices to identify the server in the network. Domain Name System (DNS) service by the DNS server helps the client to translate an unknown URL to an IP address and also cache it in the device itself for subsequent access. The local cache maintains DNS record having a map between URL and IP address.

In its most basic operation, *nslookup* tool allows the host running the tool to query any specified DNS server for a DNS record. To accomplish this task, *nslookup* sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.



```
C:\WINDOWS\system32\cmd.exe - nslookup
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

Z:\>nslookup
Default Server:  sarveshwari.amrita Vidya.edu
Address:  192.168.0.1

> saktimayi
Server:  sarveshwari.amrita Vidya.edu
Address:  192.168.0.1

Name:    saktimayi.amrita Vidya.edu
Address:  192.168.0.3

> amritapuri.org
Server:  sarveshwari.amrita Vidya.edu
Address:  192.168.0.1

Non-authoritative answer:
Name:    amritapuri.org
Address:  69.163.206.61

> hotmail.com
Server:  sarveshwari.amrita Vidya.edu
Address:  192.168.0.1

Non-authoritative answer:
Name:    hotmail.com
Addresses:  64.4.20.186, 64.4.20.169, 64.4.20.174, 64.4.20.184

>
```

The IP address of abc.com is _____ 192.155.99.34 _____

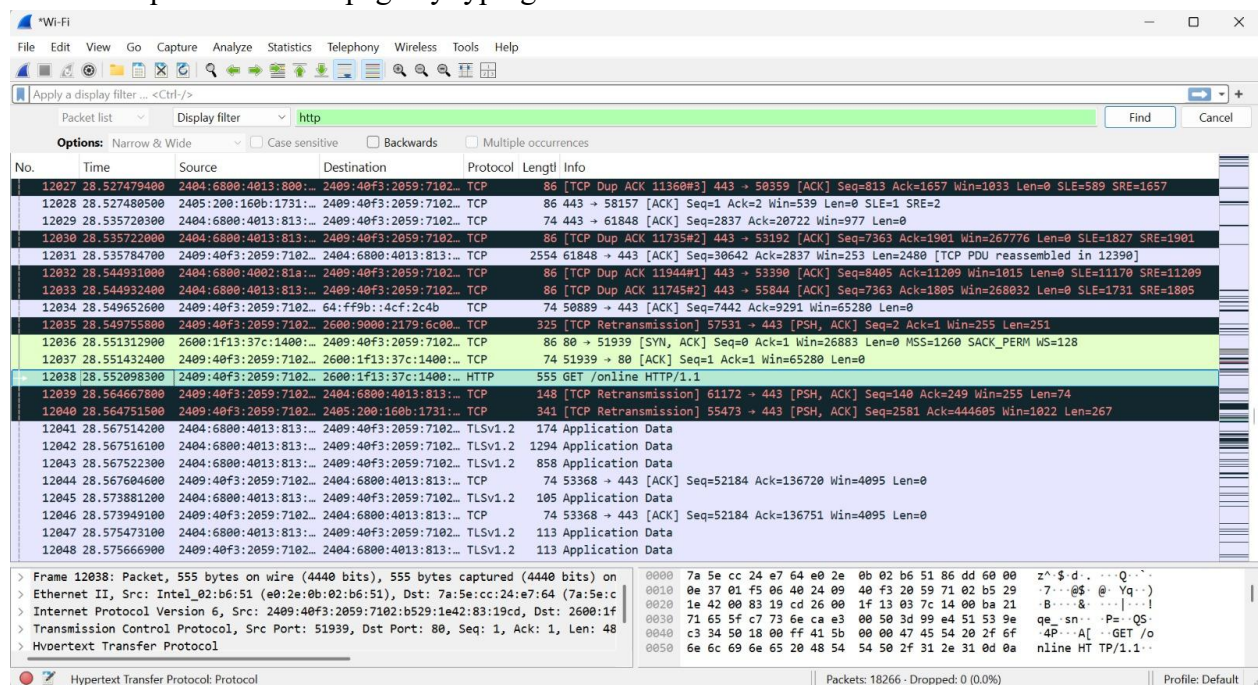
The IP address of google.com is _____ 192.178.177.102 _____

Find the IP address of all websites that you are using for Part2.

Website	IPv4 Address
Wikipedia.org	103.102.166.224
youtube.com	142.250.134.93
github.com	20.207.73.82
amazon.in	3.253.168.43

Part2: Understanding the TCP/IP protocol stack using Wireshark

1. Open Packet sniffer [Wireshark] Application and Capture the Wi-Fi/ Ethernet Interface
2. Do this activity and capture frames.
 - a. Request for a web page by typing the least used URL in the webserver



3. Briefly explain the **Encapsulation** process in at least one http request frame of the protocol analyzed. Also complete the table below by PDU contents and details requested in each layer. Also try to provide proof in the form of relevant screenshot.

Layer	Protocol	Important Contents (You can get details on clicking on a packet)	Purpose of the content specified in the Layer
Application	http	GET /, Host: neverssl.com	Requests the webpage from the server
Transport	TCP	Source Port, Destination Port, Sequence Number	Reliable data delivery
Network	IPv4	Source IP, Destination IP	Routes packets across networks
Data Link	Ethernet II	Source MAC, Destination MAC	Communication within the local network

```

Wireshark - Packet 12038 - Wi-Fi

▼ Frame 12038: Packet, 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits) on interface \Device\NPF_{4429338D-DAFF-4AD6-B74E-42C382DD8D2}, id 0
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{4429338D-DAFF-4AD6-B74E-42C382DD8D2})
    Interface name: \Device\NPF_{4429338D-DAFF-4AD6-B74E-42C382DD8D2}
    Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: Jul 26, 2026 05:18:11.809656200 India Standard Time
    UTC Arrival Time: Jul 25, 2026 23:48:11.809656200 UTC
    Epoch Arrival Time: 1785023291.809656200
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 665.900 microseconds]
    [Time delta from previous displayed frame: 665.900 microseconds]
    [Time since reference or first frame: 28.552098300 seconds]
    Frame Number: 12038
    Frame Length: 555 bytes (4440 bits)
    Capture Length: 555 bytes (4440 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ipv6:tcp:http]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
  ▼ Ethernet II, Src: Intel_02:b6:51 (e0:2e:0b:02:b6:51), Dst: 7a:5e:cc:24:e7:64 (7a:5e:cc:24:e7:64)
    ▼ Destination: 7a:5e:cc:24:e7:64 (7a:5e:cc:24:e7:64)
      .... 1. .... = LG bit: Locally administered address (this is NOT the factory default)
      .... 0. .... = IG bit: Individual address (unicast)
    ▼ Source: Intel_02:b6:51 (e0:2e:0b:02:b6:51)
      .... 0. .... = LG bit: Globally unique address (factory default)
      .... 0. .... = IG bit: Individual address (unicast)
    Type: IPv6 (0x86dd)
    [Stream index: 0]

```

- Internet Protocol Version 6, Src: 2409:40f3:2059:7102:b529:1e42:83:19cd, Dst: 2600:1f13:37c:1400:ba21:7165:5fc7:736e
 - 0110 = Version: 6
 - 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - 0000 00... = Differentiated Services Codepoint: Default (0)
 -00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
 - 0000 0000 1110 0011 0111 = Flow Label: 0x00e37
 - Payload Length: 501
 - Next Header: TCP (6)
 - Hop Limit: 64
 - Source Address: 2409:40f3:2059:7102:b529:1e42:83:19cd
 - [Address Space: Global Unicast]
 - Destination Address: 2600:1f13:37c:1400:ba21:7165:5fc7:736e
 - [Address Space: Global Unicast]
 - [Stream index: 69]
- Transmission Control Protocol, Src Port: 51939, Dst Port: 80, Seq: 1, Ack: 1, Len: 481
 - Source Port: 51939
 - Destination Port: 80
 - [Stream index: 348]
 - [Stream Packet Number: 5]
 - [Conversation completeness: Complete, WITH_DATA (31)]
 - ..0. = RST: Absent
 - ...1 = FIN: Present
 - 1... = Data: Present
 -1.. = ACK: Present
 -1. = SYN-ACK: Present
 -1 = SYN: Present
 - [Completeness Flags: ·FDASS]
 - [TCP Segment Len: 481]
 - Sequence Number: 1 (relative sequence number)
 - Sequence Number (raw): 1033495633
 - [Next Sequence Number: 482 (relative sequence number)]
 - Acknowledgment Number: 1 (relative ack number)
 - Acknowledgment number (raw): 1402913588
 - 0101 = Header Length: 20 bytes (5)
 - Flags: 0x018 (PSH, ACK)

0040	c3 34 50 18 00 ff 41 5b 00 00 47 45 54 20 2f 6f	·4···A[··GET /o
0050	6e 6c 69 6e 65 20 48 54 54 50 2f 31 2e 31 0d 0a	nline HT TP/1.1·
0060	48 6f 73 74 3a 20 63 6c 65 61 72 6f 6c 64 66 69	Host: cl earoldfi
0070	6e 65 70 6c 61 6e 2e 6e 65 76 65 72 73 73 6c 2e	neplan.n everssl.
0080	63 6f 6d 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a	com·Con nection:
0090	20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 70 67	keep-al ive··Upg
00a0	72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d 52 65	rade-Ins ecur·Re
00b0	71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 72 2d	quests: 1·User-
00c0	41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35	Agent: M ozilla/5
00d0	2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 31	.0 (Wind ows NT 1
00e0	30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 34 29	0.0; Win 64; x64)
00f0	20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 33 37	AppleWe bKit/537
0100	2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 6b 65	.36 (KHT ML, like

Accurate ECN (tcp.flags.ecn), 1 bit

☒ Show packet bytes Layout: Vertical (Stacked) ▾

Wireshark · Packet 12038 · Wi-Fi

Transmission Control Protocol, Src Port: 51939, Dst Port: 80, Seq: 1, Ack: 1, Len: 481

Source Port: 51939
Destination Port: 80
[Stream index: 348]
[Stream Packet Number: 5]

[Conversation completeness: Complete, WITH_DATA (31)]

...0. = RST: Absent
...1 = FIN: Present
.... 1... = Data: Present
.... .1.. = ACK: Present
.... ..1 = SYN-ACK: Present
.... ...1 = SYN: Present
[Completeness Flags: ·FDASS]
[TCP Segment Len: 481]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1033495633
[Next Sequence Number: 482 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1402913588
0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

000. = Reserved: Not set
...0 = Accurate ECN: Not set
.... 0... = Congestion Window Reduced: Not set
.... .0.. = ECN-Echo: Not set
.... ..0. = Urgent: Not set
.... ...1 = Acknowledgment: Set
.... 1.. = Push: Set
.... ..0.. = Reset: Not set
....0. = Syn: Not set
....0 = Fin: Not set
[TCP Flags:AP...]

Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x415b [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0

0040 c3 34 50 18 00 ff 41 5b 00 00 47 45 54 20 2f 6f .4...A[..GET /o
0050 6e 6c 69 6e 65 20 48 54 54 50 2f 31 2e 31 0d 0a nline HT TP/1.1
0060 48 6f 73 74 3a 20 63 6c 65 61 72 6f 6c 64 66 69 Host: cl earoldfi
0070 6e 65 70 6c 61 6e 2e 6e 65 76 65 72 73 73 6c 2e neplan.n everssl.
0080 63 6f 6d 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a com..Con nectio:
0090 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 70 67 keep-al ive..Upg
00a0 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d 52 65 rade-Ins ecure-Re
00b0 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 72 2d quests: 1..User-
00c0 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 Agent: Mozilla/5
00d0 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 31 .0 (Wind ows NT 1
00e0 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 34 29 0.0; Win 64; x64)
00f0 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 33 37 AppleWe bKit/537
0100 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 6b 65 .36 (KHT ML, like

Accurate ECN (tcp.flags.oe), 1 bit

Show packet bytes Layout: Vertical (Stacked)

Labsheet-2

3) Briefly explain the Encapsulation process in atleast one http requests frame of the protocol analyzed. Also complete the table below by PDU contents and details requests in each layer. Also try to provide proof in the form of screenshot.

Ans:- When a user requests a webpage, the browser generates an HTTP GET request at the application layer. The transport layer encapsulates this request into a TCP segment by adding TCP headers such as source and destination ports. The Network layer further encapsulates it into an IP packet by adding Source and destination IP addresses. Finally, the Data link layer adds source and destination MAC addresses to create an Ethernet frame, which is transmitted over the physical network medium.